

# Adrian Dar T. Serapio

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Google Scholar • LinkedIn

## Research Interests

Building embodied agents grounded in multimodal clinical data and implementing AI-driven clinical workflows validated through prospective clinical trials.

## Education

UC Berkeley-UCSF, Berkeley & San Francisco, CA 2025–Present

Ph.D. in Bioengineering, Designated Emphasis in Computational Precision Health

Advisors: Yang Yang, Kang Wang, and Jae Ho Sohn

UC Berkeley, Berkeley, CA 2020–2024

B.S. in Electrical Engineering and Computer Sciences (GPA: 3.78/4.00)

## Research Experience

Graduate Researcher, Yang Lab, Sohn Lab — UCSF 2025–Present

- Lead the development of EHRGym, a benchmark for training and evaluating autonomous computer-use agents within clinical workflows.
- Architect a multi-agent system and an OMOP-compliant environment to evaluate database cohort curation multi-task performance including early pancreatic cancer detection and HCC lesion tracking.
- Design and coordinate prospective randomized clinical trials investigating large language models for diagnostic error detection and an EHR-grounded vision-language model for contextualized imaging interpretation.

Staff Research Associate, Sohn Lab — UCSF 2024–2025

- Directed a multi-reader clinical study evaluating whether LLM-generated imaging indications, synthesized from clinical notes, improve diagnostic utility compared to referring clinician-provided indications.
- Supervised undergraduate researchers in building a patient-facing vision-language agent for lung cancer screening and developing automated evaluation metrics for patient-centered radiology report communication.

Undergraduate Researcher, Sohn Lab — UCSF 2021–2024

- Led the pipeline development of an open-source, fine-tuned LLM designed to generate radiology report impressions from the radiology report findings and a multi-reader study to assess model performance against radiologists.

Undergraduate Researcher, Data-Intensive Development Lab — UC Berkeley 2020–2021

- Designed geospatial and statistical visualizations (choropleth maps, parity plots) of machine learning-based wealth estimates in Togo, which were featured in a *Nature* cover article.

## Work Experience

Software Development Engineering Intern, Amazon Summer 2023

- Developed an observability pipeline for Ring Data Management to attribute infrastructure costs to internal services by aggregating time-series logs into daily and monthly cost estimates.

## Selected Publications

- **Serapio, A.\***, Chen, T.L.\*, Tangsombatvisit, B., Fields, B.K.K., Yu, Y., Guo, Y., Kim, S.K., Miao, B.Y., Sushil, M., Hess, C.P., Majumdar, S., & Sohn, J.H. Radiologically Relevant Clinical History Summarization with Large Language Models: A Multi-Reader Performance Study. *Radiology*, In Revision.
- **Serapio, A.**, Chaudhari, G., Savage, C.H., Lee, Y.J., Vella, M., Sridhar, S., Schroeder, J.L., Liu, J., Yala, A., & Sohn, J.H. An Open-Source Fine-Tuned Large Language Model for Radiological Impression Generation: A Multi-Reader Performance Study. *BMC Medical Imaging*, 24:254, 2024.

## Technical Skills

**Programming:** Python, JavaScript, R, Bash

**Frameworks & Tools:** PyTorch, Hugging Face, NumPy, Pandas, Matplotlib, React, Node.js, Next.js, Git, Linux, AWS, HPC/Slurm